

Zubair Mughal

LIVE CODING CLASSES FOR KIDS, TEENS & BEGINNERS

GEN

COURSE 06 OF 12
CURRICULUM GUIDE

Generative AI for Teens

Understanding and using AI responsibly

GRADE BAND

Grade 8–12

DIFFICULTY

Beginner–Intermediate

DURATION

8 weeks · 2 sessions/week

ABOUT THIS COURSE

Course Overview

A practical, responsible introduction to generative AI — how large language models and image generators actually work, how to prompt them effectively, and how to build useful AI-powered mini-projects.

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Prerequisites: None. Prior Python exposure is helpful but not required.

LEARNING OUTCOMES

What Students Will Learn

- Understand what generative AI is and how it differs from traditional software
- Write clear, effective prompts for text and image generation
- Understand the basics of how large language models are trained, and their limits
- Use AI tools responsibly, including recognizing bias and misinformation risks
- Integrate a generative AI API into a simple Python or web project
- Evaluate AI output critically rather than accepting it at face value

WHAT WE'LL USE

Tools & Technologies

ChatGPT / Claude-style tools An LLM API Python or simple HTML/JS

CURRICULUM

Module-by-Module Breakdown

MODULE	FOCUS
01 · What Is Generative AI?	Language models, image models, and how they differ from search.
02 · Prompting Fundamentals	Clarity, context and iteration.

MODULE	FOCUS
03 · Advanced Prompting Techniques	Examples, roles and step-by-step prompts.
04 · Responsible & Ethical AI Use	Bias, misinformation, academic honesty, privacy.
05 · AI Tools in Practice	Writing, coding, image and productivity tools.
06 · Connecting to an AI API	A first programmatic AI request.
07 · Building a Mini AI Project	An AI-powered tool or chatbot.
08 · Capstone Showcase	Presenting a small AI-powered creation.

LEARNING BY BUILDING

Projects Students Will Build

- AI Prompt Portfolio — A curated set of effective prompts with explanations.
- Simple AI Chatbot / Writing Assistant — Built on a real API.
- Capstone AI Mini-Project — A small AI-powered tool solving a real problem.

BEYOND THE SYNTAX

Skills Developed

Prompt engineering Critical evaluation of AI output API basics Responsible AI use

PORTFOLIO OUTCOME

A working AI-powered mini-project, plus a short written reflection on responsible AI use.

RECOMMENDED NEXT STEP

Agentic AI for Teens, or Data Science for Teens